Product Demo

Group 8

Maintenance Triage Copilot

Predict failures. Categorize them. Prioritize what to fix today.

Track 1: Predictive Model Track - Predict Early, Decide Better

Team Name: 404 AI Found

Members:

- Serene Chan Jia En
- Low Pei Shan
- Puah Ying Wen

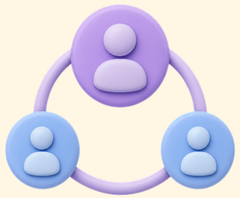
Campus: Taylor's University

Use Case

Maintenance Triage Copilot Use Case

David, Maintenance Supervisor

Every morning, David asks: “Which machine do I fix today?”



6 technicians per shift



150 machines

Option A: Fix things when they break

About RM 20,000 per breakdown in downtime, scrap and penalties.

Option B: A fixed schedule

Wastes technician time on healthy machines and misses non-tool failures.
Existing tools say “something is abnormal” - but not what, how urgent, or what to do.

Option C: Open the Triage Copilot

It predicts which machines are about to fail, names the type of failure, and ranks them by how much each fix saves, so technicians only work on the machines that matter today.



Our question: Can AI turn raw sensor data into a ranked, explainable repair schedule?

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How It Works

The Pipeline

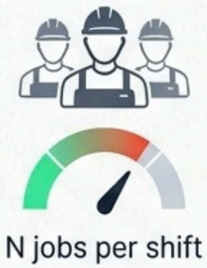


Stage	What It Does	How
Detect	Failure probability per machine	LightGBM on AI4I 2020 dataset
Categorize	Which failure mode?	Multi-class head: Tool Wear, Heat Dissipation, Power, Overstrain, Random
Prioritize	Rank by business impact	Risk Score = P(failure) x (Downtime Cost + Repair Cost)
Explain	Plain-English recommendation	SHAP values → LLM generates a one-line action

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The Prioritization Engine

Risk = P(failure) × (Downtime Cost + Repair Cost)



N jobs per shift

		FIX FIRST	
1	L-47		8.32
2	M-12		7.60
3	K-03		7.60
4	P-08		6.86
5	R-22		6.63

Output

Clean triage checklist

- [x]
- [x]
- [ ]
- [ ]

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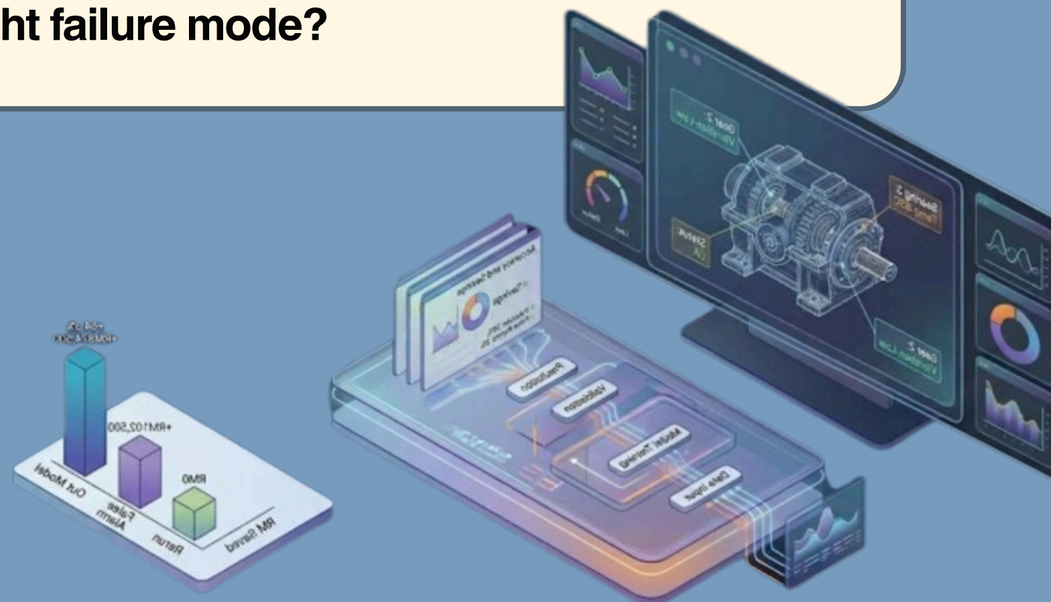
Proof & Validation

Backtest on held-out test set - 3 policies compared:

Policy	RM Saved	False Alarm Rate
Run-to-Failure	Baseline (0)	-
Fixed Schedule	+7.5% (RM102,500)	85% (149 of 175 jobs)
Our Model	+64.3% (RM874,500)	30% (25 of 82 jobs)

What we measure:

- **RM saved vs. run-to-failure and fixed-schedule baselines**
- **False alarm rate: because trust dies with noise**
- **Categorization accuracy: Did we name the right failure mode?**



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Thank You

Let's Connect and Start Building Your Future